

AUDITING GENERALIZATION CLAIMS FOR LLM AGENT HARNESSES: SEMANTIC BINDING, EXTERNAL VALIDITY, AND EVALUATION RELIABILITY

Anonymous authors

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ABSTRACT

We introduce a *claim-scope framework* for evaluating harness interventions in LLM agents, separating four progressively stronger generalization claims—local effect, within-family held-out confirmation, cross-model transfer, and cross-family transfer—and showing that evidence at one level does not imply the next. On a controlled tool-grounded *semantic-handoff* task, a non-answer-bearing “clarity bundle” (BundleS) suppresses a specific tool-green semantic-binding failure for Qwen3.7-Max (Level 0) and survives a pre-frozen held-out instance (Level 1). Under exact counterbalancing the effect is not established for DeepSeek-V4-Pro (Level 2), and on two structurally independent task families it is a clean null or a non-discriminative ceiling (Level 3). Expanding an external family prospectively from three to twelve frozen instances reversed the descriptive direction while still not establishing a treatment effect. We introduce a second, orthogonal axis—a four-layer *Harness Effect Audit Protocol* (capability, sampling, execution, artifact integrity)—and show that each layer caught a concrete threat that would have changed the scientific conclusion. An external taxonomy scope probe (Terminal-Bench 2.0→2.1, 26 repairs) shows the taxonomy has external relevance but incomplete coverage (1 direct, 21 partial, 4 outside; sampling layer 0). Our controlled task instances test existence, recurrence, and transfer direction rather than estimate population-level pass rates.

1 INTRODUCTION

When a harness intervention changes agent behavior, what can we claim? We introduce a **claim-scope ladder** with four levels and show that surviving one level does not imply surviving the next:

- **Level 0 (local effect):** a harness intervention changes behavior for a model within a controlled task family.
- **Level 1 (within-family confirmation):** the effect survives a pre-frozen held-out instance from the same semantic/generator family.
- **Level 2 (cross-model transfer):** the effect survives under a different model backend with matched task semantics.
- **Level 3 (cross-family transfer):** the effect survives on independently constructed semantic task families.

Evidence at level k does not imply evidence at level $k + 1$. Failure at a higher level does *not* refute the lower-level effect; it bounds the claim’s scope.

Our frozen evidence maps to the ladder as follows: Level 0 (Qwen3.7-Max workflow Base 1/3 → BundleS 3/3), Level 1 (pre-frozen held-out: 1/3 → 3/3), Level 2 (DeepSeek-V4-Pro: 3/4 = 3/4, not established), Level 3 (prospective STA $n=12$: not established; SPICE: ceiling/non-discriminative).

A harness-effect claim has two independent questions: (A) *what scope of generalization does the evidence support?* and (B) *was the measurement itself valid?* The claim-scope ladder addresses (A);

a four-layer audit protocol (capability, sampling, execution, artifact integrity) addresses (B). Figure 1 places our concrete evidence and incidents into this 2D framework.

Contributions. (1) A claim-scope framework separating local effects, within-family confirmation, cross-model transfer, and cross-family transfer, with the principle that each level requires independent evidence. (2) A controlled semantic-binding study showing a real local/within-family effect but no established higher-level transfer, including a prospective 12-instance expansion whose descriptive direction reversed relative to a 3-instance pilot. (3) A measurement-validity framework showing how capability, sampling, execution, and artifact failures materially changed scientific interpretation. (4) Supporting external taxonomy evidence from Terminal-Bench repairs.

2 RELATED WORK AND PROBLEM FORMULATION

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent (prompt, visible files, disclosure bundle, public tool feedback, action surface). A *harness effect* is a change in semantic-binding correctness attributable to harness information content, holding the hidden truth, grader, and tool fixed.

Problem formulation. Each task is a *semantic handoff*: bind a tuple to canonical typed roles from role-misleading evidence. The workflow family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request–authority join.

TAB vs. semantic binding. Task alignment (Mavali et al., 2026) asks *which* environmental evidence an agent should follow; semantic binding asks *which semantic role* selected evidence is permitted to fill. Our failures can occur after relevant evidence has been retrieved and can remain tool-green—the agent finds the right artifact but binds its value to the wrong typed axis.

Positioning. Yao et al. (2026) broadly characterize model $\times$ harness interactions; Wang et al. (2026) critique harness-leaderboard conflation; Ursekar et al. (2026) optimize harnesses; Zhu et al. (2025) and Shao et al. (2026) address benchmark-construction and protocol-validity standards. We differ by introducing a claim-scope ladder, isolating a controlled semantic-binding mechanism, and auditing which measurement failures changed the scientific conclusion (Table 1). Ma et al. (2026) report cross-benchmark harness gains, which we use as evidence that some interventions do transfer—so transfer must be demonstrated rather than assumed.

Table 1: Positioning relative to related work.

Prior work	What it establishes	What this paper additionally does
Harness-Bench (Yao et al., 2026)	Broad model $\times$ harness characterization	Claim-scope ladder + controlled semantic mechanism
Rethinking (Wang et al., 2026)	Leaderboard search/overfitting	Level k need not imply level $k+1$
HarnessOpt-Bench (Ursekar et al., 2026)	Optimizer capability	Studies validity of harness-effect claims
ABC (Zhu et al., 2025) / Protocol validity (Shao et al., 2026)	Setup; shortcut/reward validity	Per-control causal threat log

3 METHODOLOGY: THE 2D FRAMEWORK

The claim-scope ladder (x -axis). Any harness-effect claim lives at one of four levels (Level 0–3 above). Moving right on the ladder requires evidence at the new level; passing a lower level is necessary but not sufficient.

The measurement-validity axis (y -axis). Any measurement has four independent validity layers: (1) capability validity (does the task measure the intended capability, including tool-green semantic alternatives?); (2) sampling validity (repeated trajectories, counterbalancing, membership arbitration); (3) execution validity (transport, terminal-vs-recovered, tool health, protocol completion); (4) artifact integrity (action-surface isolation, canonical-source integrity, custody).

Figure 1: The 2D framework: claim scope (x) $\times$ measurement validity (y). Each cell lists the study’s concrete evidence or incident.

	Local	Within-family	Cross-model	Cross-family
L1 Cap.	tool-green wrong binding; typed oracle	held-out: BundleS 3/3	DeepSeek 3/4=3/4	STA $n=12$ null; SPICE ceiling
L2 Samp.	position-balanced blocking; non-det.	pre-frozen held-out inst.	counterbalanced block	prospective 12-inst. schedule
L3 Exec.	SSE streaming (censoring caught)	recovered degradation kept	—	SPICE action-surface repair
L4 Art.	canonical-tree guard	custody hashes	—	—

Construct: what semantic correctness means. Semantic correctness is *not* inferred from tool success, a hidden numeric answer, or the artifact score. It is decided solely by whether the submitted tuple matches the binding attested by the independent provenance/authority oracle. *Worked example (Family A):* visible evidence attests `intent=functional_close`, `partition=core`, `check_mode=setup`; the agent submits (`functional_close`, `core`, `both`)—PrimeTime returns green, the value is type-valid, but the coverage authority attests *setup*, not *both*; the oracle rejects it as a role-conditioned value-selection failure. This construct is instantiated independently in the workflow family (PVT axes), Family A (`intent/partition/check_mode`), and Family B (`corner/load/metric`), each with its own grader. We do *not* claim human validation (Study B was preregistered but unexecuted).

Conditions. **Base** (ambiguous handoff; no answer-bearing disclosure); **BundleS** (canonical labels, value-domain definitions, glossary, procedural contract; no answer-bearing C6; no golden values); **TypedContract** (same information as BundleS as a JSON Schema). Hidden truth and grader are identical across conditions.

Design and models. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Instance is the primary unit; repetitions nested. We evaluate Qwen3.7-Max and DeepSeek-V4-Pro; these are not representative of the frontier-model population. The design is diagnostic, not population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes.

4 STUDY I — LOCAL AND WITHIN-FAMILY EFFECT (RQ1; LEVELS 0–1)

On the workflow controlled pair (identical truth/grader; ambiguous vs. clear handoff), BundleS suppresses the axis-binding failure for Qwen3.7-Max: Base 1/3 (2 axis-binding) $\rightarrow$ BundleS 3/3 (0 axis-binding). The schema/contract subset (excluding answer-bearing C6) suffices on the development pair, and generalizes to a *pre-frozen held-out* instance: BundleS 3/3 vs Base 1/3. The failure taxonomy (Figure 2) partitions the workflow ledger (54 episodes) into 29 correct, 20 *axis-binding*, and 5 *role-conditioned value-selection*—all tool-green, rejected only by the typed oracle.

A sequential decomposition tested whether a stable minimal component underlies BundleS. None was isolated: C1 alone, C2-only, C4-only, and a C24 bridge all failed their thresholds. The bundle-level effect is real within-family; the operative sub-component is not identified.

Level 0 and Level 1 are established for Qwen3.7-Max. We make no universal improvement claim. Failure at Level 2 or Level 3 does not refute this local/within-family effect.

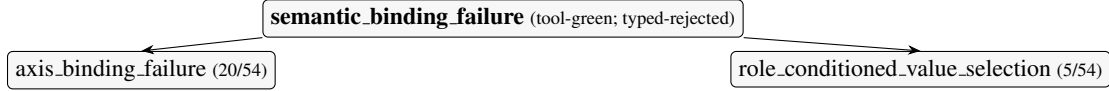


Figure 2: Semantic-binding failure taxonomy.

5 STUDY II — CROSS-MODEL AND CROSS-FAMILY TRANSFER (RQ2; LEVELS 2–3)

Does the Level 0–1 effect transfer? Table 2 is the main result.

Table 2: Main result matrix (semantic-binding correct rate; instance is the unit).

Family / model			Base	BundleS	TypedContract	<i>n</i>	verdict
Workflow	/	Qwen3.7-Max	1/3	3/3	—	1	local effect
(Level 0)							
Workflow	/	Qwen3.7-Max	1/3	3/3	—	1	within-family
(Level 1, held-out)							
Workflow	/	DeepSeek-V4-Pro	3/4	3/4	—	1	not established
(Level 2)							
STA / Qwen3.7-Max (Level 3, prospective)			0.208	0.333	0.458	12	not established
SPICE / Qwen3.7-Max (Level 3)			1.00	1.00	1.00	3	ceiling

Level 2 (cross-model). Under exact counterbalancing, BundleS is not established for DeepSeek-V4-Pro (3/4 = 3/4). This does not refute the Level 0–1 effect for Qwen3.7-Max; it bounds the claim’s scope.

Level 3 (cross-family, STA). Across 12 pre-registered instances, BundleS vs Base: 3 improving, 2 declining, 7 tied; exact sign-test $p = 1.0$; permutation sensitivity $p = 0.31$. *The three-instance pilot and the prospectively frozen twelve-instance batch differed even in their descriptive ordering. We interpret this as evidence that very small task panels are unreliable for directional Harness claims, not as evidence that BundleS transfers.* TypedContract is descriptively highest (0.458) but no preregistered TypedContract advantage was established; this is hypothesis-generating only. The pilot ($n=3$) is not pooled with the prospective $n=12$.

Table 3: Prospective STA per-instance outcomes ($n=12$; Y=correct, n=wrong).

inst.	2-rep (rate)			diff		
	Base	BundleS	TC	S–B	TC–B	TC–S
0004	nY(.5)	nn(0)	nn(0)	–.5	–.5	.0
0005–0009	nn(0)	nn(0)	nn(0)	.0	.0	.0
0010	nY(.5)	YY(1)	YY(1)	.5	.5	.0
0011	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0012	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0013	YY(1)	YY(1)	YY(1)	.0	.0	.0
0014	nn(0)	nn(0)	Yn(.5)	.0	.5	.5
0015	nn(0)	nn(0)	YY(1)	.0	1.0	1.0

Level 3 (cross-family, SPICE). Base = BundleS = TypedContract = 1.00—a non-discriminative ceiling. This is a measurement limitation, not evidence against transfer.

Three outcomes, kept distinct. The transfer evidence yields: (i) *null on a discriminating family* (STA, $n=12$): informative but cannot rule out a small effect; (ii) *ceiling* (SPICE): Base already solves the binding; no headroom; (iii) *direction reversal under prospective expansion* (STA pilot vs.

prospective): evidence that very small panels are unreliable for directional claims. We report each separately and do not pool.

6 STUDY III — MEASUREMENT VALIDITY AND EXTERNAL SCOPE PROBE (RQ3)

Each headline conclusion could have been manufactured or masked by a measurement artifact. The four-layer protocol is instantiated and stress-tested; Table 4 lists incidents where a control *materially changed the interpretation*.

Table 4: Audit-protocol incidents: each control changed the scientific conclusion.

Threat	Control	Conclusion that would have been wrong
Non-streaming censoring (L3)	SSE + terminal-transport arbiter	Qwen3.7-Max falsely recorded as failing
Position confounding (L2)	Exact position-balanced blocking	Base/BundleS order leaked into contrast
Recovered degradation (L3)	Terminal-valid vs recovered split	Hard-but-recovered solve discarded
PT corrupted measurement (L3)	Tool-health sentinel + bookends	Tool hiccup read as model failure
SPICE action-surface false positive (L4)	Immutable core + forensic audit	12 SPICE episodes zeroed (false ceiling)
Canonical-source mutation (L4)	Exact-commit worktree + hash guard	Golden rewrite read as tool outage

Why the controls are causal. Without SSE streaming, Qwen3.7-Max would have been recorded as failing the controlled pair—inverting RQ1. Without the SPICE action-surface repair, all twelve SPICE episodes would have been zeroed, manufacturing a false ceiling. Without the canonical-hash guard, a silent golden rewrite would have masqueraded as a day-long tool outage. Each would have altered the headline interpretation.

External taxonomy scope probe (Terminal-Bench 2.0→2.1). As an external scope probe (not validation), we coded the 26 officially documented repairs under the four-layer-plus-Other rubric, *frozen before coding*: **1 direct, 21 partial, 4 outside** (of 26); layers: capability 17, sampling 0, execution 10, artifact 1; 6 multi-layer. We interpret this as evidence that the taxonomy has *external relevance but incomplete coverage*. This is static retrospective taxonomy application; it does *not* validate runtime portability and does *not* validate every audit layer (L2 sampling received no external support). The full 26-task coding is in the Appendix.

7 LIMITATIONS AND DISCUSSION

Controlled executable tasks, not sampled production incidents. *EDA-heavy environments*—transfer results are scoped to EDA. *Small instance counts*—Level 0 and Level 2 rest on $n=1$; the design is diagnostic, not population-estimating. *Construct validity*—rests on executable provenance/authority oracles; the preregistered human construct-validity study (Study B) is preregistered but unexecuted; we do not claim human validation. *Model scope*—two models, not representative of the frontier population. *Audit protocol*—instantiated and stress-tested here, not broadly validated; the external probe shows relevance but incomplete coverage. *No mechanism isolated*—the bundle-level effect is real within-family but no stable minimal component was identified.

Discussion. The claim-scope ladder makes generalization claims auditable: a reader can locate any harness-effect claim at a specific level and ask whether the evidence supports it. The measurement-validity axis makes the measurement independently checkable. Together they answer: *what scope does the evidence support, and was the measurement itself valid?* Some harness interventions do transfer (Ma et al., 2026); transfer is therefore an empirical property that must be demonstrated, not assumed.

Reproducibility. Deterministic generators (fixed seeds); independent per-family graders; exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching. Program totals: 58+24+36+72 paid primary episodes; committed-ledger cost ¥745.29.

8 CONCLUSION

We introduced a claim-scope framework for harness-effect generalization claims and showed that a local/within-family effect (Levels 0–1) did not establish cross-model (Level 2) or cross-family (Level 3) transfer. A prospective 12-instance expansion reversed the descriptive direction while still not establishing a treatment effect. The measurement-validity framework showed that capability, sampling, execution, and artifact controls each materially changed the scientific conclusion. The contribution is a framework for auditing what evidence supports a generalization claim—not a proposal of an improved harness.

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A STUDY I — PER-INSTANCE LEDGER AND MINIMAL-COMPONENT ABLATION

Workflow ledger (54 episodes): 29 correct, 20 axis-binding, 5 role-conditioned value-selection. Minimal-component ablation (exploratory): no stable sub-component isolated (C1 alone, C2-only, C4-only, C24 bridge all failed thresholds).

B STUDY II — PROSPECTIVE STA DETAIL

The per-instance table (Table 3) appears in the main text. Historical pilot ($n=3$) not pooled.

C STUDY III — FULL 26-TASK TERMINAL-BENCH REPAIR CODING

Rubric frozen before coding; source: PR #53 “Terminal-Bench 2.1” (body sha256 c7172fc6; files sha256 fccf2d61); 2.0/2.1 snapshots frozen (89 tasks each; sha256 5c7ed05c).

Task	threat	mechanism	coverage
adaptive-rejection-sampler	L1 cap	instruction clarification	partial
build-pmars	L1 cap	instruction clarification; verifier/test tightening	partial
caffe-cifar-10	L3 exec; L1 cap	timeout; verifier/test broadening; instruction; env/dep; resource	partial
compile-compcert	L3 exec	environment/dependency repair	partial
configure-git-webserver	L1 cap	oracle repair; verifier/test tightening	partial
extract-moves-from-video	L3 exec	external-dependency removal	partial
filter-js-from-html	L1 cap; L3 exec	instruction clarification; resource increase	partial
fix-git	L4 art	hidden-artifact removal; verifier/test tightening	direct
hf-model-inference	L1 cap	verifier/test broadening	partial
install-windows-3.11	L1 cap	instruction clarification; verifier/test tightening	partial
make-doom-for-mips	L3 exec	oracle repair; env/dep repair	partial
mteb-leaderboard	L3 exec; L1 cap	env/dep; instruction; resource	partial
mteb-retrieve	L3 exec; L1 cap	env/dep; instruction clarification	partial
polyglot-c-py	L1 cap	verifier/test broadening	partial
polyglot-rust-c	L1 cap	verifier/test broadening	partial
protein-assembly	L1 cap	oracle repair	partial
rstan-to-pystan	L1 cap	oracle repair	partial
sam-cell-seg	L1 cap	instruction clarification	partial
torch-tensor-parallelism	L1 cap; L3 exec	instruction clarification; resource increase	partial
torch-pipeline-parallelism	L3 exec	resource increase	partial
query-optimize	L1 cap	instruction clarification	partial
train-fasttext	L3 exec; L1 cap	verifier/test tightening; env/dep repair	partial
build-pov-ray	Other	env/dep repair	not-covered
financial-document-processor	Other	env/dep repair	not-covered
mcmc-sampling-stan	Other	env/dep repair	not-covered
overfull-hbox	Other	env/dep repair	not-covered

Aggregate: direct 1 / partial 21 / not-covered 4; L1 17, L3 10, L4 1, Other 4; L2 0; 6 multi-layer.

D OPERATIONAL INDEPENDENCE AND PHASE CHRONOLOGY

Five pre-registered structural criteria (independent templates, vocabularies, truths, graders, decoys). Historical chronology (provenance; does not structure the main paper): discovery → transport repair → controlled pair → held-out → cross-model → cross-family → TypedContract → prospective STA. Episodes 58/24/36/72; costs ¥682.25 / ¥7.72 / ¥11.75 / ¥43.56.

E INFRASTRUCTURE + CUSTODY

Exact-commit worktree; canonical-hash guard (FAILED_INTEGRITY stop); episode arbiter; SSE streaming; tool-health sentinel; immutable-core/derived-deck repair; per-episode custody byte-matching. Pre-flight caught a grading-path bug (malformed signoff JSON) before the prospective run; historical pilot unaffected (0/30 triggers).

Ethics statement. The preregistered blinded human construct-validity study (Study B) is preregistered but unexecuted because qualified independent human annotators were unavailable; no LLM annotator was substituted.

AI-use statement (outside the page limit). An LLM-based assistant was used for orchestration code, generators, infrastructure, report scripts, and prose drafting. All scientific results are produced by committed deterministic pipelines reproduced from frozen manifests; no claim was authored or validated by an LLM as an oracle.

Reproducibility statement. Deterministic generators, independent graders, frozen manifests, schedules, custody hashes, and sanitized episode evidence are in the anonymized supplement. ICLR 2027 deadlines: abstract September 18, 2026 AOE; full paper September 25, 2026 AOE.